OPERATIG SYSTEMS

Introduction



What is an Operating System?

User-centric definition

- A program that acts as an intermediary between a user of a computer and the computer hardware
- Defines an interface for the user to use services provided by the system
- Provides a “view” of the system to the user
- Converts what the hardware gives to what the user wants
- The view can hide many details of the hardware that the user does not need to know
- Can even give a very different view of the operating environment to the user than what is actually there

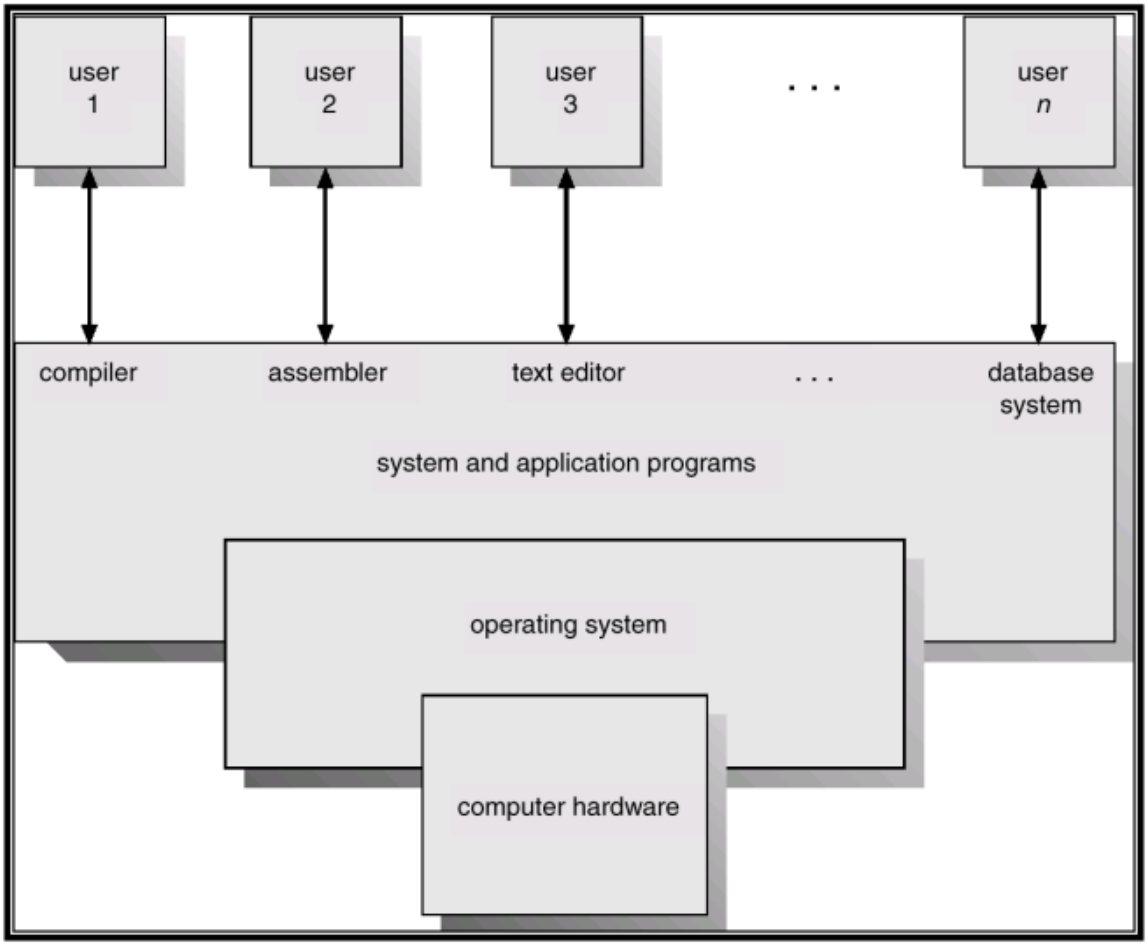
System-centric definition

- Efficiently manages and allocates resources to users
- Controls the execution of user programs and operations of I/O devices
- Provides isolation/protection between different user programs

Computer System Components

1. Hardware – provides basic computing resources (CPU, memory, I/O devices)
2. Operating system – controls and coordinates the use of the hardware among the various application programs for the various users
3. Applications programs – define the ways in which the system resources are used to solve the computing problems of the users (compilers, databases, games …)
4. Users (people, machines, other computers).

Abstract view of system components



Types of Systems

i. Batch Systems-Multiple jobs, but only one job in memory at one time and executed (till completion) before the next one starts
ii. Multi-programmed Batch Systems-Multiple jobs in memory, CPU is multiplexed between them. CPU-bound vs I/O bound jobs
iii. Time-sharing Systems-Multiple jobs in memory and on disk, CPU is multiplexed among jobs in memory, jobs swapped between disk and memory. Allows interaction with users

iv. Multiprocessing Systems
    - More than one CPU in a single machine to allocate jobs to
    - Symmetric Multiprocessing, NUMA machines …
    - Multicore

    Other Parallel Systems, Distributed Systems, Clusters…

    Different types of systems with multiple CPUs/Machines

v. Real Time Systems-Systems to run jobs with time guarantees

vi. Many other types-Embedded systems, mobiles/smartphones…

OS design depends on the type of system it is designed for

Challenges today

- Very wide variety of systems
- From small embedded systems with low memory and storage to very large systems with hundreds of thousands of machines with large distributed storage (clouds, clusters)
- Increasing number of cores per processor and processors per machine
- Virtualization

Our focus in this course:

- Uniprocessor, time-sharing systems running general purpose jobs from users-Addresses most of the core OS design issues
- Effect of multicore/multiprocessors-Most processors are multicore, many machines have more than one processor, will evaluate what additional issues this brings

Resources Managed by OS

- ➢ Physical-CPU, Memory, Disk, I/O Devices like keyboard, monitor, printer
- ➢ Logical-Process, File, …

Main Components of an OS

Resource-Centric View

- ✓ Process Management
- ✓ Main Memory Management
- ✓ File Management
- ✓ I/O System Management
- ✓ Secondary Storage Management
- ✓ Security and Protection System
- ✓ Networking (this is now integrated with most OS, but will be covered in the Networks course)

User-centric view

- ✓System Calls
- ✓Command Interpreter (not strictly a part of an OS)

Process Management

A process is a program in execution-Needs certain resources to accomplish its task-CPU time, memory, files, I/O devices…

OS responsibilities

- ✓Process creation and deletion
- ✓Process suspension and resumption

  Provide mechanisms for:
  - process synchronization
  - inter-process communication

Main-Memory Management

OS responsibilities

- ✓Keep track of which parts of memory are currently being used and by whom
- ✓Decide which processes to load when memory space becomes available
- ✓Allocate and deallocate memory space as needed
- ✓Protect memory of one process from another

File Management

OS responsibilities

- ✓File creation, deletion, modification
- ✓Directory creation, deletion, modification
- ✓Support of primitives for manipulating files and directories
- ✓Mapping files onto secondary storage.
- ✓File backup on stable (nonvolatile) storage media

I/O System Management

The I/O system consists of:

- ✓A buffer-caching system
- ✓Device driver interface
- ✓Drivers for specific hardware devices

Secondary-Storage Management

Most modern computer systems use disks as the principle on-line storage medium, for both programs and data.

OS responsibilities

- ✓Free space management
- ✓Storage allocation
- ✓Disk scheduling

Security and protection of system

Protection refers to a mechanism for controlling access by programs, processes, or users to both system and user resources.

The protection mechanism must:

- ✓distinguish between authorized and unauthorized usage
- ✓specify the controls to be imposed
- ✓provide a means of enforcement

System Calls

System calls provide the interface between a running program and the OS-Think of it as a set of functions available to the program to call (but somewhat different from normal functions, we will see why).

Generally available as assembly-language instructions.

Most common languages (e.g., C, C++) have APIs that call system calls underneath

Passing parameters to system calls

- Pass parameters in registers
- Store the parameters in a table in memory, and the table address is passed as a parameter in a register
- Push (store) the parameters onto the stack by the program, and pop off the stack by operating system

Command-Interpreter System

Strictly not a part of OS, but always there-the shell

Allows user to give commands to OS, interprets the commands and executes them-Calls appropriate functions/system calls